

CSS3 aspect-ratio Complete Guide

CSS Educational Series

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Chapter 1

Introduction to aspect-ratio

1.1 What is aspect-ratio?

The CSS `aspect-ratio` property defines a preferred aspect ratio for a box. It ensures that responsive elements maintain their proportions as they scale, making it perfect for images, videos, iframes, and flexible layouts without needing JavaScript or padding-bottom hacks.

1.1.1 Key Facts

- Maintains aspect ratio as element resizes
- Works with width/height constraints
- Supports all numeric aspect ratios (16/9, 4/3, 1/1, etc.)
- Eliminates layout shift when images load
- Prevents cumulative layout shift (CLS)
- Perfect for responsive media
- Modern browser support: Chrome 88+, Firefox 89+, Safari 15+
- No JavaScript required

1.1.2 Common Aspect Ratios

Ratio	Use Case	Example
16/9	Video, widescreen	YouTube, films
4/3	Standard display	Classic monitors
1/1	Square images	Instagram, portraits
3/2	Photos	Photography, 35mm film
21/9	Ultrawide	Gaming, cinema
9/16	Vertical video	TikTok, Stories
2/1	Wide banner	Headers, hero sections

1.1.3 Browser Support

Browser	Chrome	Firefox	Safari	Edge
aspect-ratio	88+	89+	15+	88+

Chapter 2

Basic Syntax and Values

2.1 Fundamental Syntax

```
1  /* Numeric ratio */
2  img {
3      aspect-ratio: 16/9;
4  }
5
6  /* As decimal */
7  .video {
8      aspect-ratio: 1.777; /* 16/9 */
9  }
10
11 /* Square */
12 .thumbnail {
13     aspect-ratio: 1;
14 }
15
16 /* Vertical video */
17 .story {
18     aspect-ratio: 9/16;
19 }
20
21 /* Landscape photos */
22 .gallery {
23     aspect-ratio: 3/2;
24 }
25
26 /* Auto - element's natural ratio */
27 .responsive {
28     aspect-ratio: auto;
29 }
30
31 /* Auto with preference */
32 .image {
33     aspect-ratio: auto 16/9;
34 }
```

Listing 2.1: Basic aspect-ratio Syntax

2.2 Aspect Ratio Values

```

1 <style>
2   /* Fractional notation (preferred) */
3   .ratio-16-9 {
4       aspect-ratio: 16/9;
5   }
6
7   /* Decimal notation */
8   .ratio-decimal {
9       aspect-ratio: 1.5; /* 3/2 */
10  }
11
12  /* Common ratios */
13  .square {
14      aspect-ratio: 1;
15  }
16
17  .golden-ratio {
18      aspect-ratio: 1.618; /* ~1.6/1 */
19  }
20
21  .cinemascope {
22      aspect-ratio: 21/9;
23  }
24
25  /* Auto - use intrinsic ratio */
26  .auto-ratio {
27      aspect-ratio: auto;
28  }
29
30  /* Auto with fallback ratio */
31  .auto-with-fallback {
32      aspect-ratio: auto 1920 / 1080;
33  }
34
35  /* Width-first declaration */
36  .width-first {
37      aspect-ratio: 1920 / 1080;
38  }
39 </style>
40
41 <!-- Examples applying these ratios -->
42 
43 <div class="square"></div>
44 <iframe class="ratio-16-9"></iframe>

```

Listing 2.2: Different Ratio Formats

2.3 How aspect-ratio Works with Sizing

```

1 <style>

```

```
2      /* Width defined: height calculated from ratio */
3      .video-container {
4          width: 100%;
5          aspect-ratio: 16/9;
6          /* Height becomes 56.25% of width */
7      }
8
9      /* Height defined: width calculated from ratio */
10     .portrait {
11         height: 500px;
12         aspect-ratio: 9/16;
13         /* Width becomes 281.25px */
14     }
15
16     /* Both width and height: aspect-ratio is ignored */
17     .fixed {
18         width: 400px;
19         height: 300px;
20         aspect-ratio: 16/9; /* This is ignored */
21     }
22
23     /* Min/max constraints with aspect-ratio */
24     .responsive {
25         width: 100%;
26         max-width: 1200px;
27         aspect-ratio: 16/9;
28     }
29
30     /* With padding override */
31     .container {
32         width: 100%;
33         aspect-ratio: 1;
34         padding: 20px; /* Reduces available space */
35     }
36 </style>
```

Listing 2.3: Interaction with Width and Height

Chapter 3

Practical Applications

3.1 Responsive Video Embeds

```
1 <style>
2     /* Video wrapper */
3     .video-wrapper {
4         position: relative;
5         width: 100%;
6         max-width: 800px;
7         margin: 0 auto;
8     }
9
10    /* 16:9 aspect ratio */
11    .video-container {
12        width: 100%;
13        aspect-ratio: 16/9;
14        background-color: #000;
15        border-radius: 8px;
16        overflow: hidden;
17    }
18
19    /* iframe inside container */
20    .video-container iframe {
21        width: 100%;
22        height: 100%;
23        border: none;
24    }
25
26    /* Alternative: background video */
27    .video-bg {
28        width: 100%;
29        aspect-ratio: 16/9;
30        background: url('video-bg.mp4') center / cover no-repeat;
31        background-size: cover;
32    }
33
34    /* Full-width header video */
35    .header-video {
36        width: 100%;
```

```
37     aspect-ratio: 16/9;
38     object-fit: cover;
39   }
40 </style>
41
42 <!-- YouTube embed example -->
43 <div class="video-wrapper">
44   <div class="video-container">
45     <iframe src="https://www.youtube.com/embed/..."
46       allowfullscreen></iframe>
47   </div>
48 </div>
49
50 <!-- Responsive video element -->
51 <video class="header-video" controls>
52   <source src="video.mp4" type="video/mp4">
53 </video>
```

Listing 3.1: Video Container with aspect-ratio

3.2 Image Gallery

```
1 <style>
2   .gallery {
3     display: grid;
4     grid-template-columns: repeat(auto-fit, minmax(200px, 1fr));
5     gap: 16px;
6   }
7
8   .gallery-item {
9     width: 100%;
10    aspect-ratio: 1; /* Square images */
11    overflow: hidden;
12    border-radius: 8px;
13    box-shadow: 0 2px 8px rgba(0,0,0,0.1);
14  }
15
16  .gallery-item img {
17    width: 100%;
18    height: 100%;
19    object-fit: cover; /* Maintain aspect ratio */
20    transition: transform 0.3s ease;
21  }
22
23  .gallery-item:hover img {
24    transform: scale(1.05);
25  }
26
27  /* Different ratios */
28  .gallery-wide {
29    grid-template-columns: repeat(auto-fit, minmax(250px, 1fr));
30  }
```

```
31
32 .gallery-wide .gallery-item {
33     aspect-ratio: 3/2; /* Landscape ratio */
34 }
35
36 .gallery-portrait {
37     grid-template-columns: repeat(auto-fit, minmax(150px, 1fr));
38 }
39
40 .gallery-portrait .gallery-item {
41     aspect-ratio: 9/16; /* Portrait ratio */
42 }
43
44 /* Featured image larger */
45 .gallery-featured {
46     grid-column: span 2;
47     grid-row: span 2;
48 }
49
50 .gallery-featured {
51     aspect-ratio: 1; /* Keep same ratio */
52 }
53 </style>
54
55 <div class="gallery">
56     <div class="gallery-item">
57         
58     </div>
59     <div class="gallery-item">
60         
61     </div>
62     <div class="gallery-item gallery-featured">
63         
64     </div>
65     <!-- More items -->
66 </div>
```

Listing 3.2: Gallery with Consistent Proportions

3.3 Product Grid

```
1 <style>
2     .product-grid {
3         display: grid;
4         grid-template-columns: repeat(auto-fill, minmax(200px, 1fr));
5         gap: 20px;
6     }
7
8     .product-card {
9         background: white;
10        border-radius: 8px;
11        overflow: hidden;
```

```
12     box-shadow: 0 2px 12px rgba(0,0,0,0.1);
13     transition: transform 0.2s, box-shadow 0.2s;
14 }
15
16 .product-card:hover {
17     transform: translateY(-4px);
18     box-shadow: 0 4px 16px rgba(0,0,0,0.15);
19 }
20
21 .product-image {
22     width: 100%;
23     aspect-ratio: 1; /* Square product images */
24     background: #f5f5f5;
25     display: flex;
26     align-items: center;
27     justify-content: center;
28     overflow: hidden;
29 }
30
31 .product-image img {
32     width: 100%;
33     height: 100%;
34     object-fit: cover;
35 }
36
37 .product-info {
38     padding: 16px;
39 }
40
41 .product-name {
42     font-weight: 600;
43     margin-bottom: 8px;
44     color: #333;
45 }
46
47 .product-price {
48     font-size: 18px;
49     font-weight: bold;
50     color: #d32f2f;
51     margin-bottom: 12px;
52 }
53
54 .product-button {
55     width: 100%;
56     padding: 10px;
57     background: #0066cc;
58     color: white;
59     border: none;
60     border-radius: 4px;
61     cursor: pointer;
62     font-weight: 600;
63 }
64
65 .product-button:hover {
```

```
66         background: #0052a3;
67     }
68 </style>
69
70 <div class="product-grid">
71     <div class="product-card">
72         <div class="product-image">
73             
74         </div>
75         <div class="product-info">
76             <div class="product-name">Product Name</div>
77             <div class="product-price">\$29.99</div>
78             <button class="product-button">Add to Cart</button>
79         </div>
80     </div>
81     <!-- More products -->
82 </div>
```

Listing 3.3: E-Commerce Product Layout

Chapter 4

Advanced Techniques

4.1 Maintaining Aspect Ratio with Padding

```
1 <style>
2   /* OLD METHOD: padding-bottom hack */
3   .video-old {
4     position: relative;
5     padding-bottom: 56.25%; /* 16:9 ratio */
6     height: 0;
7   }
8
9   .video-old iframe {
10    position: absolute;
11    top: 0;
12    left: 0;
13    width: 100%;
14    height: 100%;
15  }
16
17  /* NEW METHOD: aspect-ratio (much simpler) */
18  .video-new {
19    width: 100%;
20    aspect-ratio: 16/9;
21  }
22
23  .video-new iframe {
24    width: 100%;
25    height: 100%;
26  }
27
28  /* Comparison: aspect-ratio is cleaner, better performance */
29 </style>
30
31 <!-- Old way (still works, but not recommended) -->
32 <div class="video-old">
33   <iframe src="..."></iframe>
34 </div>
35
36 <!-- New way (recommended) -->
```

```
37 <div class="video-new">
38   <iframe src="..."></iframe>
39 </div>
```

Listing 4.1: Aspect-Ratio vs Padding-Bottom Hack

4.2 Responsive Aspect Ratios

```
1 <style>
2   .responsive-container {
3     width: 100%;
4     aspect-ratio: 1; /* Mobile: square */
5   }
6
7   @media (min-width: 768px) {
8     .responsive-container {
9       aspect-ratio: 16/9; /* Tablet: widescreen */
10    }
11  }
12
13  @media (min-width: 1024px) {
14    .responsive-container {
15      aspect-ratio: 21/9; /* Desktop: ultrawide */
16    }
17  }
18
19  /* Alternative: CSS custom properties */
20  :root {
21    --aspect-ratio: 1;
22  }
23
24  .container {
25    aspect-ratio: var(--aspect-ratio);
26  }
27
28  @media (min-width: 768px) {
29    :root {
30      --aspect-ratio: 16/9;
31    }
32  }
33 </style>
```

Listing 4.2: Different Ratios at Different Breakpoints

4.3 Aspect Ratio with min/max

```
1 <style>
2   /* Maintain 16:9 but constrain width */
3   .video-constrained {
4     width: 100%;
```

```
5      max-width: 1200px;
6      aspect-ratio: 16/9;
7      margin: 0 auto;
8  }
9
10 /* Maintain 1:1 with size limits */
11 .image-square {
12     width: 100%;
13     max-width: 500px;
14     min-width: 200px;
15     aspect-ratio: 1;
16 }
17
18 /* Fluid within constraints */
19 .gallery-item {
20     width: 100%;
21     max-width: 400px;
22     aspect-ratio: 3/2;
23 }
24
25 /* Size calculated from aspect ratio */
26 .auto-size {
27     width: 100%;
28     aspect-ratio: 1920 / 1080;
29     max-width: 1920px;
30 }
31 </style>
```

Listing 4.3: Constrained Aspect Ratio

Chapter 5

Layout Patterns

5.1 Hero Section with Video

```
1 <style>
2   .hero {
3     width: 100%;
4     aspect-ratio: 16/9;
5     position: relative;
6     overflow: hidden;
7     background: linear-gradient(135deg, #667eea 0%, #764ba2 100%);
8   }
9
10  .hero video,
11  .hero img {
12    width: 100%;
13    height: 100%;
14    object-fit: cover;
15    position: absolute;
16    top: 0;
17    left: 0;
18  }
19
20  .hero-content {
21    position: absolute;
22    top: 50%;
23    left: 50%;
24    transform: translate(-50%, -50%);
25    text-align: center;
26    color: white;
27    z-index: 2;
28  }
29
30  .hero h1 {
31    font-size: 48px;
32    margin: 0 0 16px 0;
33    text-shadow: 0 2px 8px rgba(0,0,0,0.3);
34  }
35
36  .hero p {
```

```
37     font-size: 20px;
38     text-shadow: 0 1px 4px rgba(0,0,0,0.3);
39 }
40
41 /* Overlay */
42 .hero::before {
43     content: '';
44     position: absolute;
45     top: 0;
46     left: 0;
47     right: 0;
48     bottom: 0;
49     background: rgba(0,0,0,0.3);
50     z-index: 1;
51 }
52
53 @media (max-width: 768px) {
54     .hero {
55         aspect-ratio: 4/3;
56     }
57
58     .hero h1 {
59         font-size: 32px;
60     }
61 }
62 </style>
63
64 <section class="hero">
65     <video autoplay muted loop>
66         <source src="hero.mp4" type="video/mp4">
67     </video>
68     <div class="hero-content">
69         <h1>Welcome</h1>
70         <p>Amazing Content Here</p>
71     </div>
72 </section>
```

Listing 5.1: Full-Width Hero with Aspect Ratio

5.2 Masonry Layout

```
1 <style>
2     .masonry {
3         display: grid;
4         grid-template-columns: repeat(auto-fill, minmax(200px, 1fr));
5         gap: 16px;
6         auto-rows: 200px;
7     }
8
9     .masonry-item {
10         overflow: hidden;
11         border-radius: 8px;
```

```
12     position: relative;
13 }
14
15 .masonry-item img {
16     width: 100%;
17     height: 100%;
18     object-fit: cover;
19 }
20
21 /* Make some items larger */
22 .masonry-item:nth-child(3n) {
23     grid-column: span 2;
24     grid-row: span 2;
25 }
26
27 /* Overlay info */
28 .masonry-item-info {
29     position: absolute;
30     bottom: 0;
31     left: 0;
32     right: 0;
33     background: linear-gradient(transparent, rgba(0,0,0,0.8));
34     color: white;
35     padding: 16px;
36     transform: translateY(100%);
37     transition: transform 0.3s ease;
38 }
39
40 .masonry-item:hover .masonry-item-info {
41     transform: translateY(0);
42 }
43 </style>
44
45 <div class="masonry">
46     <div class="masonry-item">
47         
48         <div class="masonry-item-info">
49             <h3>Image Title</h3>
50         </div>
51     </div>
52     <!-- More items -->
53 </div>
```

Listing 5.2: Masonry Grid with Mixed Aspect Ratios

Chapter 6

Real-World Examples

6.1 YouTube-Style Thumbnail Grid

```
1 <style>
2   .video-grid {
3     display: grid;
4     grid-template-columns: repeat(auto-fill, minmax(320px, 1fr));
5     gap: 16px;
6     padding: 16px;
7   }
8
9   .video-tile {
10    cursor: pointer;
11    border-radius: 8px;
12    overflow: hidden;
13    transition: box-shadow 0.2s;
14  }
15
16  .video-tile:hover {
17    box-shadow: 0 8px 16px rgba(0,0,0,0.2);
18  }
19
20  .video-thumbnail {
21    width: 100%;
22    aspect-ratio: 16/9;
23    background: #000;
24    position: relative;
25    overflow: hidden;
26  }
27
28  .video-thumbnail img {
29    width: 100%;
30    height: 100%;
31    object-fit: cover;
32  }
33
34  .video-play-icon {
35    position: absolute;
36    top: 50%;
```

```
37     left: 50%;
38     transform: translate(-50%, -50%);
39     width: 68px;
40     height: 48px;
41     background: rgba(255,255,255,0.9);
42     border-radius: 50%;
43     display: flex;
44     align-items: center;
45     justify-content: center;
46     opacity: 0;
47     transition: opacity 0.2s;
48 }
49
50 .video-tile:hover .video-play-icon {
51     opacity: 1;
52 }
53
54 .video-play-icon::after {
55     content: '';
56     width: 0;
57     height: 0;
58     border-left: 12px solid #1f1f1f;
59     border-top: 8px solid transparent;
60     border-bottom: 8px solid transparent;
61     margin-left: 4px;
62 }
63
64 .video-duration {
65     position: absolute;
66     bottom: 8px;
67     right: 8px;
68     background: rgba(0,0,0,0.8);
69     color: white;
70     padding: 4px 8px;
71     border-radius: 2px;
72     font-size: 12px;
73     font-weight: 600;
74 }
75
76 .video-info {
77     padding: 12px;
78     background: white;
79 }
80
81 .video-title {
82     font-weight: 600;
83     color: #030303;
84     margin-bottom: 4px;
85     line-height: 1.4;
86     display: -webkit-box;
87     -webkit-line-clamp: 2;
88     -webkit-box-orient: vertical;
89     overflow: hidden;
90 }
```

```
91
92     .video-meta {
93         font-size: 12px;
94         color: #606060;
95     }
96 </style>
97
98 <div class="video-grid">
99     <div class="video-tile">
100         <div class="video-thumbnail">
101             
102             <div class="video-play-icon"></div>
103             <div class="video-duration">12:34</div>
104         </div>
105         <div class="video-info">
106             <div class="video-title">Video Title Goes Here</div>
107             <div class="video-meta">Channel Name • 1.2M views • 2 weeks ago</div>
108         </div>
109     </div>
110     <!-- More videos -->
111 </div>
```

Listing 6.1: Video Thumbnail Layout

6.2 Instagram-Style Grid

```
1 <style>
2     .instagram-grid {
3         display: grid;
4         grid-template-columns: repeat(auto-fill, minmax(200px, 1fr));
5         gap: 4px;
6         padding: 4px;
7         background: #fafafa;
8     }
9
10    .post-item {
11        position: relative;
12        cursor: pointer;
13        overflow: hidden;
14        aspect-ratio: 1; /* Perfect square */
15        background: #e1e8ed;
16    }
17
18    .post-item img {
19        width: 100%;
20        height: 100%;
21        object-fit: cover;
22        transition: transform 0.3s ease, opacity 0.3s ease;
23    }
24
25    .post-item:hover img {
```

```
26     transform: scale(1.05);
27     opacity: 0.8;
28 }
29
30 .post-overlay {
31     position: absolute;
32     top: 0;
33     left: 0;
34     right: 0;
35     bottom: 0;
36     background: rgba(0,0,0,0.7);
37     display: flex;
38     align-items: center;
39     justify-content: center;
40     opacity: 0;
41     transition: opacity 0.3s ease;
42 }
43
44 .post-item:hover .post-overlay {
45     opacity: 1;
46 }
47
48 .post-stats {
49     text-align: center;
50     color: white;
51 }
52
53 .post-stat {
54     font-weight: 600;
55     font-size: 18px;
56 }
57
58 .post-stat-label {
59     font-size: 12px;
60     opacity: 0.8;
61 }
62
63 .post-icon {
64     display: inline-block;
65     margin-right: 4px;
66 }
67 </style>
68
69 <div class="instagram-grid">
70     <div class="post-item">
71         
72         <div class="post-overlay">
73             <div class="post-stats">
74                 <div class="post-stat">
75                     <span class="post-icon "></span> 1,234
76                 </div>
77                 <div class="post-stat">
78                     <span class="post-icon "></span> 56
79                 </div>
```

```
80         </div>
81     </div>
82 </div>
83 <!-- More posts -->
84 </div>
85 \end }yle>
```

Listing 6.2: Square Image Grid

Chapter 7

Performance and Optimization

7.1 Cumulative Layout Shift Prevention

```
1 <style>
2   /* PROBLEM: Image loads without aspect-ratio reserved space */
3   .bad-image {
4     width: 100%;
5     /* Layout shifts when image loads */
6   }
7
8   /* SOLUTION: aspect-ratio reserves space */
9   .good-image {
10    width: 100%;
11    aspect-ratio: 16/9;
12    background: #e0e0e0; /* Placeholder color */
13  }
14
15  .good-image img {
16    width: 100%;
17    height: 100%;
18    object-fit: cover;
19  }
20
21  /* Alternative: use width/height attributes */
22  .with-dimensions img {
23    width: 100%;
24    height: auto;
25    aspect-ratio: attr(width) / attr(height);
26  }
27 </style>
28
29 <!-- HTML with width/height attributes -->
30 <!-- This helps browser calculate aspect ratio automatically -->
31 
32
33 <!-- Or CSS aspect-ratio -->
34 
```

Listing 7.1: Using aspect-ratio to Prevent CLS

7.2 Resource Hints

```
1 <style>
2   .lazy-image {
3     width: 100%;
4     aspect-ratio: 16/9;
5     background: linear-gradient(90deg, #f0f0f0 25%, #e0e0e0 50%, #f0f0f0
6       75%);
7     background-size: 200% 100%;
8     animation: loading 1.5s infinite;
9   }
10
11   @keyframes loading {
12     0% { background-position: 200% 0; }
13     100% { background-position: -200% 0; }
14   }
15
16   .lazy-image.loaded {
17     animation: none;
18     background: none;
19   }
20 </style>
21
22 <!-- Image with loading optimization -->
23 
```

Listing 7.2: Performance with aspect-ratio

Chapter 8

Browser Compatibility

8.1 Feature Detection

```
1 <style>
2   /* Modern browsers: use aspect-ratio */
3   @supports (aspect-ratio: 1) {
4     .video {
5       aspect-ratio: 16/9;
6     }
7   }
8
9   /* Fallback for older browsers */
10  @supports not (aspect-ratio: 1) {
11    .video {
12      padding-bottom: 56.25%; /* 16:9 ratio */
13      position: relative;
14      height: 0;
15    }
16
17    .video iframe {
18      position: absolute;
19      top: 0;
20      left: 0;
21      width: 100%;
22      height: 100%;
23    }
24  }
25 </style>
26
27 <script>
28   // JavaScript feature detection
29   function supportsAspectRatio() {
30     const el = document.createElement('div');
31     return CSS.supports('aspect-ratio', '1');
32   }
33
34   if (supportsAspectRatio()) {
35     console.log('aspect-ratio is supported');
36   } else {
```

```
37     console.log('aspect-ratio is not supported');  
38     // Apply polyfill or fallback  
39   }  
40 </script>
```

Listing 8.1: Detecting aspect-ratio Support

8.2 Browser Support Matrix

Browser	Version	Support	Notes
Chrome	88+	Full	Full support
Firefox	89+	Full	Full support
Safari	15+	Full	Full support
Edge	88+	Full	Full support
Opera	74+	Full	Full support
IE 11	All	No	Not supported

Chapter 9

Quick Reference

9.1 Aspect Ratio Syntax Cheat Sheet

```
1  /* Basic syntax */
2  aspect-ratio: <ratio>;
3  aspect-ratio: auto;
4  aspect-ratio: auto <ratio>;
5
6  /* Common ratios */
7  aspect-ratio: 1;           /* 1:1 (square) */
8  aspect-ratio: 16/9;       /* 16:9 (video) */
9  aspect-ratio: 4/3;        /* 4:3 (standard) */
10 aspect-ratio: 3/2;        /* 3:2 (photo) */
11 aspect-ratio: 9/16;       /* 9:16 (vertical) */
12 aspect-ratio: 1.777;      /* 16:9 as decimal */
13 aspect-ratio: auto;       /* Use intrinsic ratio */
14 aspect-ratio: auto 1920/1080; /* Prefer intrinsic, fallback to 1920/1080 */
15
16 /* With sizing */
17 width: 100%;
18 aspect-ratio: 16/9;
19
20 /* Responsive */
21 @media (max-width: 768px) {
22     aspect-ratio: 4/3;
23 }
24
25 /* In calc() */
26 width: calc(100% - 20px);
27 aspect-ratio: 16/9;
```

Listing 9.1: Complete Syntax Reference

9.2 Common Patterns

- Video containers (16/9 ratio)
- Image galleries (1/1 square)

- Social media feeds (specific ratios)
- Product grids (consistent sizing)
- Hero sections (16/9 or custom)
- Responsive embeds (YouTube, Vimeo)
- Thumbnail grids (various ratios)
- Portrait images (9/16 vertical)
- Advertisement banners (custom ratios)

9.3 Best Practices

1. Use fractional notation (16/9) instead of decimals
2. Combine with max-width for responsive sizing
3. Use with object-fit for image fitting
4. Set placeholder background color
5. Include width/height HTML attributes
6. Provide @supports fallbacks for older browsers
7. Test responsive behavior at different breakpoints
8. Use with lazy loading attributes
9. Consider CLS impact when images load
10. Document aspect ratios in design systems

Chapter 10

Advanced Use Cases

10.1 Responsive Images with Different Ratios

```
1 <style>
2   .picture-wrapper {
3     width: 100%;
4     aspect-ratio: 16/9;
5   }
6
7   .picture-wrapper img {
8     width: 100%;
9     height: 100%;
10    object-fit: cover;
11    object-position: center;
12  }
13
14  /* Different ratios per breakpoint */
15  @media (max-width: 768px) {
16    .picture-wrapper {
17      aspect-ratio: 4/3;
18    }
19  }
20
21  @media (max-width: 480px) {
22    .picture-wrapper {
23      aspect-ratio: 1/1;
24    }
25  }
26 </style>
27
28 <!-- Picture element with multiple sources -->
29 <picture>
30   <source media="(min-width: 1200px)" srcset="image-wide.jpg">
31   <source media="(min-width: 768px)" srcset="image-medium.jpg">
32   
33 </picture>
```

Listing 10.1: Picture Element with Aspect Ratios

10.2 Aspect Ratio with Grid

```
1 <style>
2   .grid-container {
3     display: grid;
4     grid-template-columns: repeat(auto-fit, minmax(250px, 1fr));
5     gap: 16px;
6   }
7
8   .grid-item {
9     aspect-ratio: 1; /* All items square */
10    background: linear-gradient(135deg, #667eea, #764ba2);
11  }
12
13  /* Variable aspect ratios */
14  .grid-item:nth-child(2n) {
15    aspect-ratio: 16/9;
16  }
17
18  /* Span multiple columns */
19  .grid-item:nth-child(3) {
20    grid-column: span 2;
21    aspect-ratio: 2/1;
22  }
23
24  /* Using auto-fit vs auto-fill */
25  .auto-fit {
26    grid-template-columns: repeat(auto-fit, minmax(200px, 1fr));
27  }
28
29  .auto-fill {
30    grid-template-columns: repeat(auto-fill, minmax(200px, 1fr));
31  }
32 </style>
```

Listing 10.2: CSS Grid with Aspect Ratios

Chapter 11

Conclusion

11.1 Summary

The CSS `aspect-ratio` property revolutionizes responsive design by providing a simple, native way to maintain aspect ratios. It eliminates the need for padding-bottom hacks, prevents cumulative layout shift, and creates better user experiences with reserved space for media elements.

11.2 Key Advantages

- Simplifies responsive design
- Prevents cumulative layout shift
- No JavaScript required
- Works with any sizing constraints
- Supports flexible and fixed layouts
- Compatible with modern design patterns
- Improves performance
- Better semantic HTML

11.3 Related Features

Explore these complementary CSS properties:

- `object-fit` - Image fitting within containers
- `object-position` - Image positioning
- `width` / `height` - Sizing constraints
- `max-width` / `max-height` - Responsive limits
- CSS Grid - Advanced layout systems
- Flexbox - Flexible layouts

11.4 Resources

- [MDN Web Docs: aspect-ratio property](#)
- [W3C CSS Sizing Module Level 4](#)
- [Web.dev: Aspect Ratio Guide](#)
- [CSS Tricks: Aspect Ratio Boxes](#)